

Aryan Kumar

[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Email](#) | [6204815914](#)

Experience

IBM Edunet Foundation

June 2024 – Aug 2024

Data Science Intern — Remote

- Conducted exploratory data analysis on customer behavior data using **Python libraries** to identify key purchase patterns.
- Built a **predictive model using Random Forest** to forecast customer churn with 86% accuracy.
- Created **interactive dashboards in Power BI** to visualize KPIs such as churn rate, average customer lifetime value, and satisfaction index.
- Collaborated with the data engineering team to automate daily data cleaning pipelines, reducing manual preprocessing time by 30%.

Projects

AWS Pipeline for YouTube Trend Analysis | AWS (S3, Glue, Athena), PySpark, QuickSight

[GitHub](#)

- Architected and built a scalable, **serverless software application** on **AWS** using **S3, Glue, and Athena**.
- Engineered an automated **ETL workflow** with **PySpark** to process over **100,000 records**, deploying it as a resilient data service.
- Integrated the backend data service with a BI tool (**QuickSight**) via **SQL APIs** to deliver key performance metrics.
- Implemented a robust data validation and **error-handling framework** in **Glue**, ensuring **99%+** processing accuracy for the application.

Amazon Review Sentiment Analysis | Python, NLP, Hugging Face

[GitHub](#)

- Formulated a **sentiment analysis model** classifying 50,000+ reviews with VADER and RoBERTa, achieving 23% higher accuracy over baseline.
- Executed text preprocessing on **50,000+ reviews** using **Pandas and NLTK**, performing tokenization, normalization, and stop-word removal to enhance model input quality.
- Generated **visualizations (8+ charts)** with Matplotlib/Seaborn to compare lexicon vs transformer-based approaches.

AI-Powered Resume and Cover Letter Generator | Python, Streamlit, LangChain, Groq LLM

[GitHub](#)

- Built an **AI-driven web app** with Streamlit and LangChain to auto-generate **ATS-friendly resumes and cover letters** for 500+ users.
- Integrated **Groq LLMs (Llama 3, Mixtral)** for contextual text generation, boosting content accuracy by **90%**.
- Deployed a secure **PDF export pipeline** with authentication and encryption, reducing data leak risk by **100%**.
- Optimized model response speed by **35%** through API call batching and efficient prompt chaining.

Cold Mail Generator (GenAI) | Python, Llama 3.1, ChromaDB, LangChain, Streamlit

[GitHub](#)

- Engineered an advanced **GenAI application** using Llama 3.1 and LangChain for generating **personalized cold emails**, achieving **92% contextual accuracy**.
- Designed a **RAG pipeline** with **ChromaDB**, improving email relevance and response rate by **75%**.
- Optimized **prompt chaining and memory components**, reducing latency by **40%** and enhancing response coherence.
- Deployed a scalable **Streamlit app** with API integration, enabling **real-time content generation** for 300+ active users.

Certifications

- Career Essentials in Data Analysis – **Microsoft & LinkedIn** [view](#)
- Amazon AWS Fundamentals – **Coursera** [view](#)

Education

SRM University, Andhra Pradesh

2022 – 2026

B.Tech in Computer Science and Engineering — CGPA: 8.06/10 (current)

Technical Skills

Programming Languages	Python (Pandas, NumPy, Scikit-learn, PySpark), SQL, C++, ReactJS
Data Engineering & Big Data	ETL/ELT Pipelines, Data Warehousing, Hadoop, DBT, Kafka, Airflow
Cloud Platforms & Services	AWS (S3, Glue, Athena, Lambda, EC2), Docker, Snowflake Basics
Databases	MySQL, PostgreSQL, MongoDB
AI & Machine Learning	Machine Learning, Deep Learning, NLP, Streamlit, Langchain
Data Visualization & BI	Power BI, Tableau, AWS QuickSight, Automation Workflows

Achievements

- Won a hackathon by designing and building **"Hirrd,"** a full-stack, scalable job portal with a secure authentication module.